A Two-Step Model for Startup Funding:
Sector-level self-exciting counts and within-sector ranking

Alberto Gennaro

24 June 2026

1 Overview

The task is to predict, from weekly funding data, which startup is funded next. The model factorises this into two layers that match the structure of the data:

1. a *sector* layer that predicts *where and when* funding events occur — a weekly multivariate self-exciting count model over M sectors, with positive lagged self- and cross-excitation and market covariates;
2. a *within-sector* layer that predicts *which* startup receives a given sector event — a conditional ranker over the live startups in that sector, with a recency ("cooldown") term that suppresses startups that funded recently.

The decomposition is exact under a marked-process view: the probability that the next event is a particular startup is the sector intensity times the within-sector choice probability. It also separates a low-dimensional, identifiable counting problem (the sector layer) from a high-cardinality, dynamic-risk-set problem (the ranker). Both layers are fit by maximum likelihood. The implementation is in `hawkes_calibration/sector_ranker.py`; the end-to-end synthetic backtest is in `sector_backtest.py` and `experiments/exp14_sector_ranker.py`. Numerical results quoted in Section 7 are from the project's learning report (five-seed synthetic backtest); no experiment is re-run here.

2 Data and notation

Time is discrete (weeks) $t = 1, \dots, T$. There are M sectors and a roster of startups that changes over time. Observed quantities:

- $Y_{s,t} \in \mathbb{N}$: number of funding events in sector s in week t.
- $X_t \in \mathbb{R}^p$: market-level covariates (e.g. a risk-on regime).
- $Z_{j,t} \in \mathbb{R}^q$: features of startup j at week t (quality, founder/investor signal, traction, a sector-momentum feature).
- $\mathcal{R}_{s,t}$: the live risk set — the startups currently in sector s and active at week t. Its size changes as startups enter or exit.
- events (t, s, i^*): in week t, sector s, startup i^* was funded.

3 Data: sources, features, and run modes

Sources. Two licensed (non-public) providers supply the inputs. *FactSet Standardized Economics* provides the macro covariates — interest rates, inflation and unemployment — via the series `FFUNDT`, `FFUND` (US Fed Funds target / effective), `CPI_RAW`, `CPI_YOY` (US CPI level / year-over-year) and `RUNEMP` (unemployment rate). *PitchBook* provides the event stream and company attributes through its Deals, Company and Investors tables. Both are subscription products and are not publicly available. The macro series, however, have free public equivalents on FRED (`FEDFUNDS` or `EFFR`, `CPIAUCSL`, `UNRATE`);[1] for deal-level PE/VC activity the closest free substitutes are the Crunchbase free tier and SEC Form D filings (public, with a short lag). Because the deal data is proprietary, the pipeline runs on synthetic data by default and reads user-supplied CSVs in a real mode.

From raw fields to model inputs. A deal's `primary_industry_sector` defines its sector s and its `deal_date` its week t; the sector count $Y_{s,t}$ in (1) is the number of deals in (s, t).

- **Sector covariates** $X_{s,t}$ (the baseline of (1)): the standardized macro series and their first differences (shared across sectors); plus per-sector weekly aggregates from the Deals table, lagged one week and standardized — total amount raised ($\sum$`deal_size`), amount per event (total / count), mean `deal_size`, the up/down/flat round ratio (`vc_round_up_down_flat`), mean `pre_money_valuation` and `post_valuation`, mean `revenue` and `ebitda`, and mean `number_of_employees`. These are known before week t, so they are valid regressors; the self- and cross-excitation themselves enter through the lagged counts in (1).
- **Startup features** $Z_{j,t}$ (the score (3)): company- and deal-level attributes — age (from `year_founded`), `total_raised_to_date`, `last_financing_size`, round/stage (`vc_round`), `number_of_employee` `revenue`, `ebitda`, `revenue_growth_since_last_deal`, `total_investor_ownership`, `business_status` and `hq_global_region`; and dynamic features — weeks since the last financing (the recency / cooldown $c^{(K)}_{j,t}$), the number of prior rounds, and a recent-sector-activity "momentum" feature. Continuous features are standardized; categoricals are one-hot encoded.
- **Risk set** $\mathcal{R}_{s,t}$: a company enters at its first appearance (or `year_founded`) and leaves when `current_financing_status` or `ownership_status` indicates an acquisition, IPO or closure; the live set in sector s at week t is the candidate pool the ranker normalises over.

Run modes. The estimation of Sections 4–5 is identical in both modes; only the source of the raw columns differs.

- *Synthetic* (default). The generator (`simulate_synthetic_startup_market`) populates the raw columns above with plausible distributions — log-normal, positively correlated deal sizes and valuations; categorical rounds; AR(1) macro series; random founding dates and entries — and the same aggregation/feature pipeline builds X, Z, Y and the risk sets. No proprietary data is required.
- *Real* (`--mode real --data-dir DIR`). Two CSV files are read from `DIR`: `factset_macro.csv` (a date column plus the macro series `FFUNDT`, `FFUND`, `CPI_RAW`, `CPI_YOY`, `RUNEMP`, in long or wide form) and `pitchbook_deals.csv` (one row per deal, with at least `company_id`, `deal_id`, `deal_date`, `primary_industry_sector`, `deal_size`, `vc_round`, `vc_round_up_down_flat`, `pre_money_valuation`

[1] https://fred.stlouisfed.org

post_valuation, total_raised_to_date, number_of_employees, revenue, ebitda, year_founded, current_financing_status, hq_global_region). Optional pitchbook_companies.csv and pitchbook_investors.csv, joined on company_id and investor_id, add further company and investor features. Preprocessing maps deal_date to a week index, primary_industry_sector to s, forward-fills the macro series onto the weekly grid, aggregates deals into the (s, t) counts and sector aggregates, and constructs the startup features and dynamic risk sets.

Feature reduction. The raw PitchBook schema is wide — dozens of deal- and company-level fields, many sparse, skewed or collinear — which inflates the variance of the ranker's sector-specific weights and invites over-fitting. Preprocessing therefore (i) log-transforms the monetary fields (deal_size, valuations, revenue), standardises continuous features, groups rare categories and adds explicit missingness indicators; and (ii) reduces dimension before fitting — a truncated SVD / PCA on the standardised continuous block keeps the top r components (those explaining, say, 90% of variance), and the sector-specific ranker weights are given a low-rank form $u_s = U_s V$ with $r \ll q$, cutting the parameter count from Mq to $(M+q)r$. This is the low-rank / embedding device used to scale relational point-process models (Know-Evolve, DyRep) and keeps the high-cardinality feature set from overwhelming the conditional-logit fit.

4 Step 1: the sector-level self-exciting count model

Model. Sector counts follow a Poisson log-linear autoregression — a discrete-time multivariate Hawkes process,

$$Y_{s,t} \sim \mathrm{Poisson}(\Lambda_{s,t}), \qquad \log \Lambda_{s,t} = a_s + \beta_s^\top X_t + \sum_{r=1}^{M} \sum_{\ell=1}^{L} b_{s,r,\ell}\, Y_{r,t-\ell}, \tag{1}$$

with intercepts a_s, covariate loadings β_s, and nonnegative lag coefficients $b_{s,r,\ell} \geq 0$. The diagonal terms $b_{s,s,\ell}$ are self-excitation (a sector that funded recently funds more); the off-diagonal terms $b_{s,r,\ell}$ $(r \neq s)$ are cross-sector contagion. An optional adjacency mask forces chosen entries to zero. The nonnegativity constraint keeps the rate positive and the excitation interpretable.

Estimation. The parameters $\theta = (a, \beta, b)$ are estimated by penalised Poisson maximum likelihood. Up to a constant the objective is

$$\min_{\theta:\, b \geq 0} \sum_{t=L+1}^{T_{\mathrm{tr}}} \sum_{s=1}^{M} \left(\Lambda_{s,t} - Y_{s,t} \log \Lambda_{s,t}\right) + \tfrac{\lambda}{2}\left(\|\beta\|^2 + \|b\|^2\right), \tag{2}$$

solved with a box-constrained quasi-Newton method (L-BFGS-B); the gradient is available in closed form, and the intercepts are left unpenalised. The problem is convex, so the fit is unique. Held-out scoring uses the one-step rate $\Lambda_{s,t}$ formed from the realised lagged counts.

Stability. Equation (1) is stable (the mean count stays finite) only if the excitation has spectral radius below one. The estimator constrains $b \geq 0$ but not the spectral radius, so a fitted model can be supercritical; in the synthetic study a fit reached spectral radius ≈ 3.7. This does not affect one-step held-out scoring, but it makes *forward simulation* explode and must be corrected (project the fitted excitation to spectral radius below one, and cap the inner event loop) before simulating paths.

5 Step 2: the within-sector survival model with an outside option

Given a sector event from Step 1, we model which startup receives it as a discrete-time competing-risks survival problem over the live risk set, with one addition that matches a positives-only deals feed: an explicit *outside* alternative — "a firm not in our dataset" — so an event that goes to a company we do not track is not forced onto an observed candidate.

Candidates and the outside option. Each tracked candidate $j \in \mathcal{R}_{s,t}$ (the live startups in sector s that we follow) has a log-hazard

$$q_{j,t} = (w_0 + u_s)^\top Z_{j,t} + \eta_s \, c_{j,t}^{(K)}, \tag{3}$$

with global feature weights w_0, sector deviations u_s, and a cooldown term ($c_{j,t}^{(K)}$ is startup j's funding count over the past K weeks; $\eta_s \leq 0$ makes a recently funded startup less likely to fund again — self-inhibition). The outside alternative 0 carries a log-hazard built from *sector-level* features only, since it has no company covariates,

$$q_{0,s,t} = b_0 + \gamma^\top g_{s,t}, \qquad g_{s,t} = \big(\log(1 + |\mathcal{R}_{s,t}|),\ \text{recent sector activity}\big). \tag{4}$$

The recipient is drawn from $\mathcal{R}_{s,t} \cup \{0\}$ by a softmax over the augmented set,

$$\Pr(i \mid s,t) = \frac{\exp(q_{i,t})}{\exp(q_{0,s,t}) + \sum_{j \in \mathcal{R}_{s,t}} \exp(q_{j,t})}, \tag{5}$$

i ranging over the tracked candidates and the outside option. Dropping q_0 recovers the plain risk-set ranker.

Why the outside option fits the data. A deals database records only firms that raised; the universe of firms that could have raised is unobserved, and company features are typically available only at a firm's last round. The outside option turns this from a defect into a modelled quantity. At training, events to firms outside the tracked universe (a watch-list / portfolio) are labelled "outside" and supply the signal for (b_0, γ); at prediction, $\Pr(\text{outside} \mid s,t)$ is the calibrated probability that the next funded firm is one we do not yet track — a useful output in itself, since a high value says "widen the watch-list." Firms never seen carry no features and are simply not candidates; they fall under the outside mass.

Estimation. The parameters $(w_0, u, \eta, b_0, \gamma)$ are estimated by conditional maximum (partial) likelihood,

$$\min_{\eta \leq 0} \sum_{(t,s,i^*)} \Big(\log \big[\exp(q_{0,s,t}) + \sum_{j \in \mathcal{R}_{s,t}} \exp(q_{j,t}) \big] - q_{i^*,t} \Big) + (\text{ridge penalties}), \tag{6}$$

with i^* the funded candidate or the outside option. The objective is convex and box-constrained (cooldown nonpositive); its gradient is the softmax residual. The estimator (`fit_startup_survival`) is numpy-only, with SciPy used when available.

Relation to standard models. Equation (6) is the log-sum-exp partial likelihood of a discrete choice over the risk set — McFadden's conditional logit *with an outside good*, equivalently the Cox proportional-hazards partial likelihood (competing risks: each candidate plus an outside cause) and the Plackett–Luce choice model. The cooldown is a time-varying covariate; the negative-constrained η_s and the self-exciting sector layer below it are what separate this model from the feature-only benchmark of Section 8. On the synthetic market the analytic gradient matches finite differences to 10^{-8}, the fitted cooldown is nonpositive, the outside option discriminates off-list events (area under ROC ≈ 0.68), and the within-universe ranking beats chance (top-5 ≈ 0.70 over $\sim$7 candidates); see `tests/test_sector_survival.py`.

6 Producing a ranked watch-list

Within a sector, startups are ranked by (5) (or by the score (3)). For an overall "who is funded next" list across sectors, the marked-process factorisation gives a propensity

$$\text{score}_{j,t} \;\propto\; \Lambda_{s(j),t} \cdot \Pr(j \mid s(j), t), \tag{7}$$

the sector intensity from Step 1 times the within-sector choice probability from Step 2. Ranking quality is measured on held-out events by the top-k hit rate (is the funded startup in the top k), the mean reciprocal rank (MRR), and the held-out negative log-likelihood, each compared against a random-over-the-risk-set baseline and against the *oracle* ranker that uses the true parameters (the best achievable given the stochastic draw).

7 Study case: a synthetic market

Data-generating process. This is the synthetic run mode of Section 3; the real mode applies the identical estimation and evaluation to user-provided FactSet/PitchBook CSVs. The simulator (`simulate_synthetic_startup_market`) mirrors the two-layer model. Sector counts are drawn from (1) with a sparse, strongly diagonal positive excitation matrix (self-excitation per sector plus two cross-sector links) and AR(1) market covariates. Each sector event is assigned to a live startup through the risk-set softmax (5) with a *negative* true cooldown, so the ground truth has both self-excitation (sectors) and self-inhibition (startups). About a quarter of startups enter after the start, so risk-set sizes change over time; startup features include a quality signal and a sector-momentum-like feature.

Protocol. A market of $M = 11$ sectors and roughly 35 startups per sector is simulated for $T = 180$ weeks; the first 120 weeks train both layers, the last 60 are held out. Reported figures are averaged over five seeds.

Results (from the project learning report). The sector layer improves the held-out one-step Poisson log-likelihood over a historical-mean baseline. The ranker, on held-out weeks with about 35 candidates per pick, attains

	risk-set ranker	oracle (true parameters)	random
top-5 hit rate	≈ 0.49	0.54	0.14
MRR	≈ 0.88 of oracle	—	—

The ranker is three to six times better than chance and recovers roughly 80–92% of the oracle across seeds. The remaining gap to the oracle is the irreducible stochasticity of the funded-startup draw over the risk set, not model error. The sector-layer spectral-radius caveat of Section 4 applies to forward simulation, not to this one-step held-out scoring.

8 Benchmark: a feature-based ranker without explicit self-excitation

The two-step model encodes self-excitation and self-inhibition as *mechanisms*: the nonnegative lagged excitation matrix in (1), and the sign-constrained cooldown coefficient in (3). A natural benchmark removes these mechanisms and instead supplies engineered *features* that proxy for the same effects, to test whether the explicit structure earns its place.

Construction. Keep the risk-set evaluation identical (same held-out events, same live risk sets, same metrics), but:

- **No sector self-excitation.** Replace Step 1 by a feature-only rate — a Poisson GLM on the covariates X_t and an exogenous trailing-activity feature (e.g. a moving average of recent sector counts treated as an input, not an estimated kernel), or simply the historical-mean rate. There is no estimated lagged excitation matrix.
- **No explicit self-inhibition.** Rank startups with a survival / discrete-time hazard model (a Cox proportional-hazards or DeepHit-style logistic-hazard ranker) in which "weeks since last funding" and a "recent sector momentum" enter as ordinary covariates with free (unconstrained) coefficients. The recency feature is meant to absorb the cooldown; the momentum feature to absorb the sector self-excitation.

This benchmark is exactly a standard survival ranker with hand-built momentum and recency features. The package already provides the survival machinery for it (`SURVIVAL_RANKING.md`, `experiments/exp16_survival.py`: Kaplan–Meier, a discrete-time hazard with a ranking loss, concordance and Recall@k).

What the comparison isolates. Run the two-step model and the feature benchmark on the same backtest and compare top-k, MRR and the time-dependent concordance index. Two readings are possible and both are informative:

- If the explicit two-step model wins, the *mechanism* (a self-exciting sector kernel; a sign-constrained, risk-set-normalised cooldown) carries information that hand-built features do not — chiefly the cross-sector contagion in (1) and the dynamic risk-set normalisation in (5), neither of which a per-startup feature reproduces.
- If the benchmark matches it, the effects are well summarised by recency and momentum features, and the simpler, cheaper survival ranker suffices.

The honest prior is that the gap on the *who* question (Step 2) is modest, because the cooldown is itself a recency feature; the larger differentiator is the *where and when* question (Step 1), where cross-sector excitation and the contagion dynamics are hard to capture with a single per-sector momentum feature.

9 Modeling assumptions and alternatives

What the data supports. The two layers make very different demands on the data. The sector layer needs only weekly counts per sector and macro covariates, both well observed; it is robust. The within-sector ranker is more demanding: it assumes that at each week the live risk set of candidate startups can be enumerated and that every candidate has features. Two facts about PE/VC data strain this. First, company covariates (revenue, valuation, employees) are typically refreshed only *at a financing event*, so between rounds they are stale and for never-funded firms they are absent — features arrive on an event-driven, irregular schedule, not continuously. Second, a deals database records *positives* (firms that raised); the universe of firms that could have raised but did not — the negatives that define the risk set — is largely unobserved.

When the model is appropriate. The conditional risk-set ranker is sound when "candidates" is a defined, feature-complete universe that we track — a watch-list, a fund's screened pipeline, or a portfolio — rather than the entire private market. There the risk set is observed, features are maintained, and the softmax over candidates is well grounded; that is the recommended deployment.

Alternatives. If only the positives-only deals table is available, or features are sparse, three reformulations fit the data better.

- **The outside-option survival model (adopted as Step 2).** The model of Section 5 is exactly this reformulation: a competing-risks partial likelihood over the tracked risk set with an explicit outside cause, so it never has to enumerate the unobserved negatives and it reports Pr(outside) directly. The natural extension is a fuller per-firm *time-to-event* version (Cox / DeepHit over weeks since the last round, with right-censoring and last-observation-carried-forward features), with the sector intensity multiplying the per-firm hazard, $\Lambda_{s,t} \cdot \text{hazard}_{j,t}$.
- **Lean on always-observable signals.** The strongest, always-computable features come from the event stream itself — recency (weeks since last round), round number, sector momentum — consistent with the empirical finding that financing recency dominates next-round prediction. A model built primarily on these, with hard covariates as optional refinements, is robust to the missingness above.
- **Use investor activity.** The Investors table is observed and informative: "which firm is funded next" is partly "which firm an active investor selects." A bipartite investor–company dynamic-graph model (Know-Evolve / DyRep style) ranks companies by predicted investor–company edge formation and avoids relying on company financials.

In short, the sector layer is well-supported and the two-step architecture is sound, but the within-sector ranker should either be scoped to a tracked candidate universe or recast as a per-firm survival model; the cold-start of never-seen firms and the unobserved negatives are the binding constraints, not the choice of estimator.

10 Limitations

- Results are on synthetic data drawn from the same two-layer family the model assumes; the real test is a back-test on observed weekly funding data.

- The sector excitation is not spectral-radius-constrained during fitting; forward simulation requires projecting to the stable region.
- The decay structure (number of lags L, cooldown window K) is fixed, not selected; the cross-excitation entries are weakly identified when sector counts are low.
- The benchmark of Section 8 is specified but not yet run; it is the recommended next experiment.

11 References

1. Rizoiu, Soen, Li, Calderon, Dong, Menon, Xie (2022). Interval-censored Hawkes processes. JMLR 23(1). https://www.jmlr.org/papers/v23/21-0917.html
2. Hawkes (1971). Spectra of some self-exciting and mutually exciting point processes. Biometrika 58(1).
3. Fokianos, Rahbek, Tjøstheim (2009). Poisson autoregression. JASA 104(488):1430–1439. (discrete-time self-exciting counts.)
4. McFadden (1974). Conditional logit analysis of qualitative choice behavior. In *Frontiers in Econometrics*. (the risk-set choice model.)
5. Cox (1972). Regression models and life-tables. JRSS-B 34:187–220. (partial likelihood over the risk set.)
6. Plackett (1975). The analysis of permutations. Applied Statistics 24(2):193–202. (ranking / choice likelihood.)
7. Kaplan & Meier (1958). Nonparametric estimation from incomplete observations. JASA 53.
8. Lee, Zame, Yoon, van der Schaar (2018). DeepHit. AAAI. https://cdn.aaai.org/ojs/11842/11842-13-15370-1-2-20201228.pdf
9. Antolini, Boracchi, Biganzoli (2005). A time-dependent discrimination index for survival data. Statistics in Medicine 24(24).

Related documents: `SURVIVAL_RANKING.md` (the survival ranker in full); `NEXT_FUNDING_DESIGN.md` (the application design); `paper/investment_case_study.pdf` (the continuous-time Hawkes/MBPP foundation).